YEAR 5 MATHS · NUMBER · PREMIUM WORKBOOK

Percentages Power-Up

Fractions, decimals and percentages made simple, with real-life problems, 3 printable games and a final challenge.

7 lessons

3 games

Final test + certificate

Name _____

Date started _____

Work through one lesson at a time. Mistakes are information, not failure. Check the answer key only after you have tried.

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Start here: your learning journey

Learning intention: I know my plan and what I already know.

Name _____ **Date** _____

Your journey map

Step	What you will learn
Lesson 1	What “per cent” means (out of 100)
Lesson 2	One amount, three names: fraction, decimal, percentage
Lesson 3	Converting and ordering
Game 1	Triple Match (cut-out cards)
Lesson 4	Finding 50%, 25%, 75% and 10% of amounts
Lesson 5	Building any percentage: 1%, 5%, 20%, 15%
Game 2	Percent Path (board game)
Lesson 6	Money smart: sales and discounts
Lesson 7	Percentages in data and tests
Game 3	Crack the Code
Challenge	Reasoning and mistakes to fix
Final test	Show what you know, then collect your certificate

Quick check: no pressure!

1. 50 out of 100 = _____ %
2. 1/4 as a decimal = _____
3. 10% of 80 = _____
4. 3/4 as a percentage = _____ %
5. Which is bigger, 0.5 or 5%? _____

How did it feel? **Easy · OK · Tricky**

Word bank

Word	Meaning
Per cent (%)	out of 100
Fraction	part of a whole, like 3/4
Decimal	a number with a point, like 0.75
Benchmark	an easy, friendly number: 10%, 25%, 50%, 75%
Discount	the amount taken off a price

Parent/Teacher tip: Read the journey map together. Ask your child which lesson they think will be easiest and which will be hardest, then revisit that answer at the end.

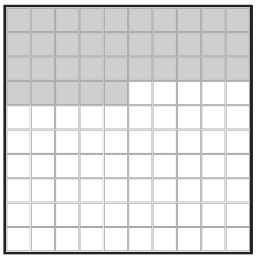
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Lesson 1: What does “per cent” mean?

Name _____ Date _____

Learning intention: I can explain “out of 100”.

Learn



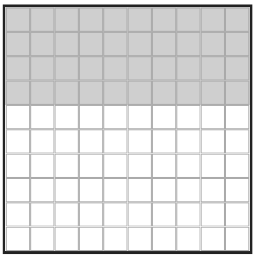
35 squares shaded out of 100

Per cent means **“out of 100”**. The symbol is **%**.

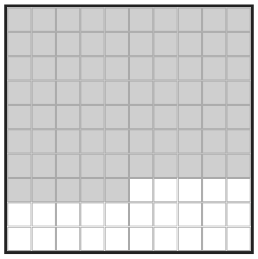
35 out of 100 = **35/100** = **0.35** = **35%**

Remember: 100 squares = 100% = the whole thing.

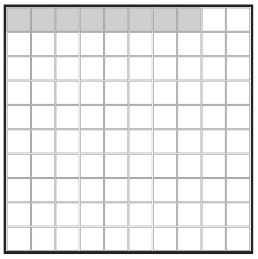
A. Write the percentage shaded



_____ %

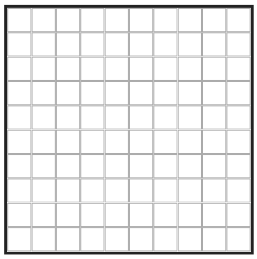


_____ %

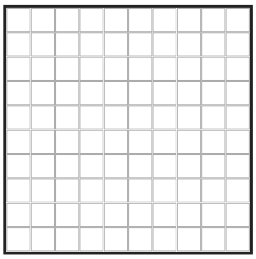


_____ %

B. Shade it yourself



Shade 60%



Shade 25%

Think: Mia says 50% means “half”. Is she right? Explain with 100 squares.

_____

_____

C. Quick fire

1. 20 out of 100 = _____ %
2. 90% means _____ out of 100
3. 100% of a pizza = _____
4. 0% means _____

Parent/Teacher tip: Ask “How many out of 100?” every time you see a % sign: sale tags, battery icons, quiz scores.

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Lesson 2: One amount, three names

Learning intention: I can write one amount three ways.

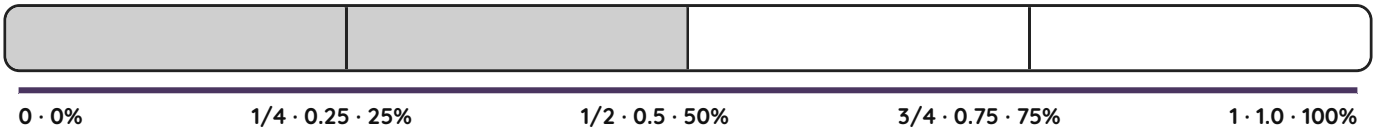
Name _____ Date _____

Learn: they are all the same

Fraction	Decimal	Percentage
3/5 = 6/10	0.6	60%
7/20 = 35/100	0.35	35%
1/4 = 25/100	0.25	25%

① Make the fraction “out of 100”. ② The top number is the percentage. ③ Write it as a decimal: 35/100 = 0.35.

Percent number line



0 · 0% **1/4 · 0.25 · 25%** **1/2 · 0.5 · 50%** **3/4 · 0.75 · 75%** **1 · 1.0 · 100%**

Place value helps: tenths and hundredths

Ones	.	Tenths	Hundredths	As a percentage
0	.	3	0	30%
0	.	0	7	7%
0	.	4	5	45%
0	.	1	5	15%

A. Complete the chain (use ×100 or ÷100)

9/10 = 90/100 = _____ = _____ %

3/20 = 15/100 = _____ = _____ %

1/5 = 20/100 = _____ = _____ %

1/50 = 2/100 = _____ = _____ %

B. Spot the pattern

What do you notice when a fraction has a denominator of 10, 20, 25 or 50? Why are they easy to turn into percentages?

_____

_____

Parent/Teacher tip: If your child is stuck, say “How many hundredths is that?” and let them double, ×4 or ×5 to reach 100.

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Lesson 3: Converting and ordering

Learning intention: I can convert and order fractions, decimals, percentages.

Name _____ **Date** _____

A. Fill in the missing parts

Fraction	Decimal	Percentage
1/2	0.5	_____
1/5	_____	20%
_____	0.75	75%
2/5	0.4	_____
1/20	_____	5%
7/10	0.7	_____
_____	0.09	9%
3/20	0.15	_____

B. Which is more?

0.3 or 3% ? _____

0.5 or 45% ? _____

1/4 or 20% ? _____

3/4 or 0.7 ? _____

C. Order from smallest to largest

Set 1: 0.6 45% 1/2 3/4 0.05

_____ < _____ < _____ < _____ < _____

Set 2: 1/4 0.35 40% 3/10

_____ < _____ < _____ < _____

D. Explain your thinking

Tip: change everything to percentages first. Write how you ordered Set 1.

_____

_____

_____

Parent/Teacher tip: Turning every number into a percentage first makes comparing much easier. Encourage the habit.

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Game 1: Triple Match

Cut out the cards, shuffle and match the triples.

Name _____ Date _____

1/4	0.25	25%
3/5	0.6	60%
1/20	0.05	5%
7/10	0.7	70%

How to play

Solo: match all 4 triples as fast as you can. Time yourself and try to beat your best.
2 players: turn all cards face down. Take turns to flip 3. A match is a triple (fraction + decimal + percentage). Keep matches and go again. Most triples wins.
Make your own: use the blank cards below to create 2 new triples.

your card	your card	your card	your card	your card	your card

Scoreboard

Round 1 time: _____ Round 2 time: _____ Round 3 time: _____

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Lesson 4: Finding 50%, 25%, 10% and 75%

Name _____ Date _____

Learning intention: I can find 50%, 25%, 10%, 75% of an amount.

Learn: percentages are fractions in disguise

Percentage	Fraction	How to find it
50%	1/2	divide by 2
25%	1/4	divide by 4
10%	1/10	divide by 10
75%	3/4	divide by 4, then × 3

Worked example: 25% of $80

$20	$20	$20	$20

25% = one quarter. $80 ÷ 4 = **$20**. So 25% of $80 is **$20**.

A. Your turn

50% of 60 = _____

25% of 40 = _____

10% of 350 = _____

75% of 20 = _____

Strategy box: use the bar to show 75% of 20.

B. Real-life problems

1. A jacket costs $90 and is 10% off. How much do you save? _____
2. Shoes cost $60 and are 50% off. What do you pay? _____
3. 20 students are in a class and 25% walk to school. How many walk? _____

C. Reason it out

Which is bigger: **25% of 80** or **10% of 200**? Show your working.

_____

_____

Parent/Teacher tip: To find 10%, move each digit one place to the right (350 → 35). Then 5% is half of that and 20% is double.

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Lesson 5: Build any percentage

Learning intention: I can build 1%, 5%, 15%, 20% from 10%.

Name _____ **Date** _____

Learn: start with 10% or 1%

30% of 80: 10% = 8, so 30% = 8 × 3 = **24**

15% of 80: 10% = 8, 5% = 4, so 15% = 8 + 4 = **12**

1% of 700: divide by 100 = **7**

Build-it bars: 15% of 60

10% = 6	**5% = 3**

15% = 10% + 5% = 6 + 3 = **9**

A. Your turn

30% of 50 = _____

15% of 40 = _____

20% of 90 = _____

60% of 20 = _____

5% of 60 = _____

1% of 300 = _____

B. Show your build

Find **35% of 120**. Fill the boxes: 10% = _____ 30% = _____ 5% = _____ 35% = _____

10%	**10%**	**10%**	**5%**

C. Problem

A bike costs $200. A shop takes 15% off. How much is the discount? What is the new price? Show your thinking.

_____

_____

_____

Parent/Teacher tip: Stress the habit: 10% first, then build. 1% and 5% are just 10% divided by 10 or by 2.

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Practice zone

Learning intention: I can find percentages and solve problems.

Name _____ **Date** _____

A. Quick calculations

1. 50% of 18 = _____
2. 25% of 36 = _____
3. 10% of 120 = _____
4. 75% of 40 = _____
5. 20% of 45 = _____
6. 15% of 60 = _____
7. 5% of 80 = _____
8. 1% of 900 = _____

B. Convert

3/4 = _____ % 0.5 = _____ % 1/10 = _____ % 17% = _____/100

C. Problem solving

1. Sam scored 18 out of 20 on a quiz. Write his score as a percentage. _____
2. A $48 book has 25% off. What is the sale price? Show your steps.

_____

3. A glass holds 200 mL. It is 75% full. How many mL of water are in it? _____

D. Find the mistake

Liam says: “10% of 90 means 90 ÷ 100, so the answer is 0.9.” What did he do wrong? What is the correct answer?

_____

E. Create your own

Write a shopping problem that uses 10% or 50%. Swap with a partner and solve it.

_____

_____

High Five Stars

I can convert fractions, decimals and percentages ☆ ☆ ☆

I can find a percentage of an amount ☆ ☆ ☆

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Game 2: Percent Path

2 players. You need a die and 2 counters (coins or rubbers).

Name _____ Date _____

1 START Place your counter here	**2** 50% of 20	**3** 25% of 12	**4** 0.3 as a %	**5** 10% of 80
10 10% of 350	**9** 1/5 as a %	**8** 75% of 8	**7** 3/4 as a %	**6** Roll again!
11 Oops! Go back 2	**12** 50% of 90	**13** 0.05 as a %	**14** 25% of 100	**15** 20% of 50
20 FINISH You made it!	**19** 15% of 20	**18** 5% of 60	**17** 1/10 as a %	**16** Go forward 2!

Rules

1. Roll the die and move your counter. **2.** Answer the question on your square out loud. Your partner checks the answer key. **3.** Correct: stay. Wrong: go back 1 square. **4.** Dashed squares are special. **5.** First to land on FINISH wins. You need the exact number!

Winner's reflection

Which question was trickiest? How did you solve it?

_____

_____

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Lesson 6: Money smart: sales and discounts

Name _____ Date _____

Learning intention: I can find discounts and sale prices.

Learn

Discount = the amount taken off. **Sale price** = original price − discount.

Example: a $60 game is 25% off. 25% of $60 = $15. Sale price = $60 − $15 = **$45**.

A. The sale table

Item	Price	Discount	You save	Sale price
Backpack	$40	25%	_____	_____
T-shirt	$24	50%	_____	_____
Book	$15	10%	_____	_____
Sneakers	$80	20%	_____	_____
Cap	$12	25%	_____	_____

B. Best buy?

Shop A: $50 with 20% off. **Shop B:** $45 with 10% off.
Which shop is cheaper, and by how much? Show your steps.

C. Design a sale

You run a shop. Choose 3 items, set a price and a discount (10%, 25% or 50%). Work out each sale price.

Item	Price	Discount	Sale price

Parent/Teacher tip: Next time you shop, let your child work out the sale price before the checkout does.

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Lesson 7: Percentages in data and tests

Name _____ Date _____

Learning intention: I can use percentages for data and scores.

Learn

When the total is 20, each person is **5%** (because 20 × 5 = 100). When the total is 10, each is **10%**. When the total is 25, each is **4%**.

A. Favourite fruit (20 students)

Apple: 8 students

Banana: 5 students

Mango: 4 students

Grapes: 3 students

Fruit	Students	Percentage
Apple	8	_____ %
Banana	5	_____ %
Mango	4	_____ %
Grapes	3	_____ %

B. Test scores

1. 17/20 = _____ %

2. 9/10 = _____ %

3. 21/25 = _____ %

C. Think

1. Which two fruits together make 45%?

_____

2. Ava got 15/20 and Ben got 80%. Who did better? How do you know?

_____

_____

Parent/Teacher tip: Use a deck of cards or counters to model “out of 20” and “out of 25” percentages.

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Game 3: Crack the Code

Solve each clue, then decode the secret word.

Name _____ **Date** _____

Decoder key

30	9	50	12	20	5	15	40
P	E	R	C	N	T	A	S

Clues

☐ 1. 50% of 60
= _____

☐ 2. 25% of 36
= _____

☐ 3. 10% of 500
= _____

☐ 4. 75% of 16
= _____

☐ 5. 50% of 18
= _____

☐ 6. 20% of 100
= _____

☐ 7. 5% of 100
= _____

Secret word

☐ ☐ ☐ ☐ ☐ ☐ ☐

Write the 7 letters in order: this word is the topic of this book!

Level up: make your own code

Pick a word of 4 to 6 letters. Give each letter an answer to a percentage question. Challenge a friend to crack it.

_____

_____

_____

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Challenge zone

Learning intention: I can reason and explain my thinking.

Name _____ **Date** _____

1. True or false? Explain.

★★ 10% of 200 is the same as 20% of 100.

★★ 50% of any number is bigger than 25% of the same number.

2. Work backwards

★★ 25% of a number is 9. What is the number? _____

★★★ 10% of a number is 7. What is the number? _____

3. Double discount

★★★ A $60 jacket is 50% off. Then the shop takes another 50% off the new price. Is it now free? Show why or why not.

4. Investigate

★★★ Find 3 different pairs of percentage + amount that give an answer of 12. (Example: 50% of 24.)

My thinking

Which challenge did I enjoy most? What strategy helped me?

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Final test

Total: 20 marks. Work on your own, then check your score.

Name _____ **Date** _____

A. Convert (4 marks)

1. 3/10 = _____ %
2. 0.45 = _____ %
3. 7/20 = _____ %
4. 60% = _____ (decimal)

B. Calculate (6 marks)

5. 50% of 74 = _____
6. 25% of 80 = _____
7. 10% of 460 = _____
8. 75% of 60 = _____
9. 20% of 35 = _____
10. 5% of 40 = _____

C. Problems (5 marks)

11. A $70 jacket is 30% off. How much do you save and what do you pay? (2 marks) _____ _____

12. Kai got 16 out of 20. What percentage is that? (1 mark) _____

13. Order smallest to largest: 0.35, 3/10, 40%, 1/4 (2 marks)
_____ < _____ < _____ < _____

D. Challenge (5 marks)

14. A phone costs $200. Store A: 25% off. Store B: $20 off, then another 10% off the new price. Which store is cheaper and by how much? (3 marks)

_____

_____

15. 20% of a number is 14. What is the number? (2 marks) _____

Score tracker

My score: _____ / 20 **17–20** Percentage Pro · **12–16** Nearly there · **0–11** Keep practising (revisit the lessons you found tricky).

Questions I got wrong:

_____

What I will do to improve:

_____

_____

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CERTIFICATE OF ACHIEVEMENT

Percentages Power-Up

This certifies that

_____

has completed the Year 5 Percentages Workbook and can find and compare fractions, decimals and percentages.

Final test score: _____ / 20

Date _____ Signed _____

My reflection

Something I am proud of:

_____

Something I want to practise more:

_____

My next goal:

_____

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Answer key: Part 1

Teacher/parent copy.

Name _____ **Date** _____

Start here: quick check

1) 50 2) 0.25 3) 8 4) 75% 5) 0.5 (it is 50%, bigger than 5%)

Lesson 1

A) 40%, 75%, 8% B) check 60 and 25 squares shaded; “Think”: yes, 50 out of 100 is half. C) 20; 90; the whole pizza; none/nothing.

Lesson 2

A) 9/10 = 0.9 = 90%; 1/5 = 0.2 = 20%; 3/20 = 0.15 = 15%; 1/50 = 0.02 = 2%. B) The denominator is a factor of 100, so you can scale up to 100 by multiplying (×10, ×5, ×4, ×2).

Lesson 3

A) 50%; 0.2; 3/4; 40%; 0.05; 70%; 9/100; 15%. B) 0.3 (30% > 3%); 1/4 (25% > 20%); 0.5 (50% > 45%); 3/4 (75% > 70%).
C) Set 1: 0.05 < 45% < 1/2 < 0.6 < 3/4. Set 2: 1/4 < 3/10 < 0.35 < 40%.

Game 1 matches

1/4 = 0.25 = 25%; 3/5 = 0.6 = 60%; 1/20 = 0.05 = 5%; 7/10 = 0.7 = 70%.

Lesson 4

A) 30; 35; 10; 15. B) 1) $9 2) $30 3) 5 students. C) Both are 20, so they are equal.

Lesson 5

A) 15; 18; 3; 6; 12; 3. B) 10% = 12; 30% = 36; 5% = 6; 35% = 42. C) 15% of $200 = $30 discount; new price $170.

Teacher notes and common errors

Using this workbook: one lesson per sitting (15–25 min), games as rewards or revision, test at the end.
Common errors: thinking 10% means “divide by 100”; forgetting to subtract the discount from the price; mixing up 25% (÷4) with 20% (÷5); thinking 0.3 equals 3%.
Support: use the 100 grid and bar models; keep numbers friendly (multiples of 10).
Extend: ask “How could you check this a different way?” and use the Challenge zone.

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Answer key: Part 2

Teacher/parent copy.

Name _____ **Date** _____

Practice zone

A) 1) 9 2) 9 3) 12 4) 30 5) 9 6) 9 7) 4 8) 9. B) 75%; 50%; 10%; 17. C) 1) 90% 2) $12 off, so $36 3) 150 mL. D) He divided by 100 instead of 10; correct answer is 9. E) Answers vary.

Game 2 answers (square: answer)

2: 10 · 3: 3 · 4: 30% · 5: 8 · 7: 75% · 8: 6 · 9: 20% · 10: 35 · 12: 45 · 13: 5% · 14: 25 · 15: 10 · 17: 10% · 18: 3 · 19: 3

Lesson 6

A) Backpack $10 / $30; T-shirt $12 / $12; Book $1.50 / $13.50; Sneakers $16 / $64; Cap $3 / $9. B) Shop A: $50 − $10 = $40. Shop B: $45 − $4.50 = $40.50. Shop A is cheaper by 50c.

Lesson 7

A) Apple 40%, Banana 25%, Mango 20%, Grapes 15%. B) 85%; 90%; 84% (21 × 4). C) 1) Banana + Mango (25% + 20% = 45%). 2) Ava 15/20 = 75%; Ben 80%, so Ben did better.

Game 3: Crack the Code

30, 9, 50, 12, 9, 20, 5 → P E R C E N T. The secret word is **PERCENT**.

Challenge zone

1) True (20 = 20). True for any positive number, because half is more than a quarter. 2) 36; 70. 3) No: $60 → $30 → $15. 4) Examples: 50% of 24, 25% of 48, 10% of 120, 20% of 60.

Final test (20 marks)

1) 30% 2) 45% 3) 35% 4) 0.6 5) 37 6) 20 7) 46 8) 45 9) 7 10) 2 11) Save $21, pay $49 (1 + 1) 12) 80% 13) 1/4 < 3/10 < 0.35 < 40% (2) 14) Store A: $150. Store B: $180, then 10% = $18, so $162. A is cheaper by $12 (1 + 1 + 1). 15) 70 (1 for 20% → 10% = 7, 1 for answer).

Progress tracker

Tick when complete and note the date.

Section	Done	Date	Notes
Lessons 1–3	☐		
Game 1	☐		
Lessons 4–5	☐		
Game 2	☐		
Lessons 6–7	☐		
Game 3	☐		
Challenge zone	☐		
Final test	☐		

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